MA1522 Linear Algebra for Computing
L2 Lesson 23



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Semester 1, 2026/27

Question 1

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Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

a. Find the QR-factorization of $\boldsymbol{A}$.

b. Find a 4×4 orthogonal matrix $\boldsymbol{Q}'$ and a 4×3 matrix $\boldsymbol{R}' = (r'_{ij})$ with $r'_{ij} = 0$ for all $i > j$, such that $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$.

Question 1a: Solution

Performing Gram-Schmidt to the columns $\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\}$ of $\boldsymbol{A}$, we get

$$\begin{aligned}
\boldsymbol{v}_1 &:= \boldsymbol{c}_1 && = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}; \\
\boldsymbol{v}_2 &:= \boldsymbol{c}_2 - \frac{\boldsymbol{c}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 && = \boldsymbol{c}_2 - \frac{2}{2}\boldsymbol{v}_1 && = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix}; \\
\boldsymbol{v}_3 &:= \boldsymbol{c}_3 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 && = \boldsymbol{c}_3 - \frac{6}{2}\boldsymbol{v}_1 - \frac{6}{3}\boldsymbol{v}_2 && = \begin{pmatrix} -1 \\ 1 \\ -2 \\ 1 \end{pmatrix}.
\end{aligned}$$

Thus $\boldsymbol{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{7}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{7}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{7}} \\ 0 & 0 & \frac{1}{\sqrt{7}} \end{pmatrix} \begin{pmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \sqrt{7} \end{pmatrix} =: \boldsymbol{QR}.$

Question 1a: Solution

Performing Gram-Schmidt to the columns $\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\}$ of $\boldsymbol{A}$, we get

$$\begin{aligned}
\boldsymbol{v}_1 &:= \boldsymbol{c}_1 && = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}; \\
\boldsymbol{v}_2 &:= \boldsymbol{c}_2 - \frac{\boldsymbol{c}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 && = \boldsymbol{c}_2 - \frac{2}{2}\boldsymbol{v}_1 && = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix}; \\
\boldsymbol{v}_3 &:= \boldsymbol{c}_3 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{c}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 && = \boldsymbol{c}_3 - \frac{6}{2}\boldsymbol{v}_1 - \frac{6}{3}\boldsymbol{v}_2 && = \begin{pmatrix} -1 \\ 1 \\ -2 \\ 1 \end{pmatrix}.
\end{aligned}$$

Thus $\boldsymbol{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{7}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{7}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{7}} \\ 0 & 0 & \frac{1}{\sqrt{7}} \end{pmatrix} \begin{pmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \sqrt{7} \end{pmatrix} =: \boldsymbol{QR}.$

Question 1b: Solution

Let $\boldsymbol{Q}' = \left(\begin{array}{c|c} \boldsymbol{Q} & \boldsymbol{w} \end{array} \right)$ be 4×4, and let $\boldsymbol{R}' = \begin{pmatrix} \boldsymbol{R} \\ \boldsymbol{0} \end{pmatrix}$ be 4×3.

Then $\boldsymbol{R}' = (r_{ij})$ with $r_{ij} = 0$ for all $i > j$ and $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$.

For $\boldsymbol{Q}'$ to be orthogonal, we need $\|\boldsymbol{w}\| = 1$ and $\boldsymbol{w} \cdot \boldsymbol{v}_i = 0$ for $i = 1, 2, 3$.

$\boldsymbol{e}_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \notin \operatorname{span}\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\} = \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, and

$$\boldsymbol{v}_4 := \boldsymbol{e}_1 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}\boldsymbol{v}_3 = \begin{pmatrix} \frac{1}{42} \\ -\frac{1}{42} \\ \frac{1}{21} \\ \frac{1}{7} \end{pmatrix}$$

is orthogonal to $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ and $\boldsymbol{v}_3$. Let $\boldsymbol{w} = \frac{1}{\|\boldsymbol{v}_4\|}\boldsymbol{v}_4$.

Thus $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$ where

$$\boldsymbol{Q}' = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{7}} & \frac{1}{\sqrt{42}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{7}} & -\frac{1}{\sqrt{42}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{7}} & \frac{2}{\sqrt{42}} \\ 0 & 0 & \frac{1}{\sqrt{7}} & \frac{6}{\sqrt{42}} \end{pmatrix}, \quad \boldsymbol{R}' = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \sqrt{7} \\ 0 & 0 & 0 \end{pmatrix}.$$

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Let $\boldsymbol{Q}' = \left(\begin{array}{c|c} \boldsymbol{Q} & \boldsymbol{w} \end{array} \right)$ be 4×4, and let $\boldsymbol{R}' = \begin{pmatrix} \boldsymbol{R} \\ \boldsymbol{0} \end{pmatrix}$ be 4×3.

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For $\boldsymbol{Q}'$ to be orthogonal, we need $\|\boldsymbol{w}\| = 1$ and $\boldsymbol{w} \cdot \boldsymbol{v}_i = 0$ for $i = 1, 2, 3$.

$$\boldsymbol{e}_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \notin \operatorname{span}\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\} = \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}, \text{ and}$$

$$\boldsymbol{v}_4 := \boldsymbol{e}_1 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 - \frac{\boldsymbol{e}_1 \cdot \boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}\boldsymbol{v}_3 = \begin{pmatrix} \frac{1}{42} \\ -\frac{1}{42} \\ \frac{1}{21} \\ \frac{1}{7} \end{pmatrix}$$

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Thus $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$ where

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For $\boldsymbol{Q}'$ to be orthogonal, we need $\|\boldsymbol{w}\| = 1$ and $\boldsymbol{w} \cdot \boldsymbol{v}_i = 0$ for $i = 1, 2, 3$.

$\boldsymbol{e}_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \notin \operatorname{span}\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\} = \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, and

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is orthogonal to $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ and $\boldsymbol{v}_3$. Let $\boldsymbol{w} = \frac{1}{\|\boldsymbol{v}_4\|}\boldsymbol{v}_4$.

Thus $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$ where

$$\boldsymbol{Q}' = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{7}} & \frac{1}{\sqrt{42}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{7}} & -\frac{1}{\sqrt{42}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{7}} & \frac{2}{\sqrt{42}} \\ 0 & 0 & \frac{1}{\sqrt{7}} & \frac{6}{\sqrt{42}} \end{pmatrix}, \quad \boldsymbol{R}' = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \sqrt{7} \\ 0 & 0 & 0 \end{pmatrix}.$$

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For $\boldsymbol{Q}'$ to be orthogonal, we need $\|\boldsymbol{w}\| = 1$ and $\boldsymbol{w} \cdot \boldsymbol{v}_i = 0$ for $i = 1, 2, 3$.

$\boldsymbol{e}_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \notin \operatorname{span}\{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\} = \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, and

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is orthogonal to $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ and $\boldsymbol{v}_3$. Let $\boldsymbol{w} = \frac{1}{\|\boldsymbol{v}_4\|}\boldsymbol{v}_4$.

Thus $\boldsymbol{A} = \boldsymbol{Q}'\boldsymbol{R}'$ where

$$\boldsymbol{Q}' = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{7}} & \frac{1}{\sqrt{42}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{7}} & -\frac{1}{\sqrt{42}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{7}} & \frac{2}{\sqrt{42}} \\ 0 & 0 & \frac{1}{\sqrt{7}} & \frac{6}{\sqrt{42}} \end{pmatrix}, \quad \boldsymbol{R}' = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} & 2\sqrt{3} \\ 0 & 0 & \sqrt{7} \\ 0 & 0 & 0 \end{pmatrix}.$$

Question 2

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As a simplified model for weather forecasting, the weather is characterized as sunny, cloudy, or rainy. Suppose that there is a 50% chance that the weather on the next day will be the same as today's weather, and a 25% chance that it will be one of the other two types of weather. Assuming the weather is sunny on day 0, let a_n, b_n, and c_n denote the probabilities that it is sunny, cloudy, or rainy on day n, respectively.

(a) Let $\boldsymbol{x}_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$. Find a symmetric matrix $\boldsymbol{A}$ such that $\boldsymbol{A}\boldsymbol{x}_n = \boldsymbol{x}_{n+1}$ for every nonnegative integer n.

(b) Find an invertible matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

(c) Use (b), or otherwise, to find the general formulas for a_n, b_n and c_n (in terms of n).

(d) In a long run (as n tends large), find the probability that a day is a sunny, cloudy and rainy day respectively.

Question 2(a,b): Solution

a. $\boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$.

b. $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x - \frac{1}{4})^2(x - 1)$.

$$\frac{1}{4}\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Basis for $E_{\frac{1}{4}} = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$.

$$1\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{2} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Basis for $E_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

Question 2(a,b): Solution

a. $\boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$.

b. $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x - \frac{1}{4})^2(x - 1)$.

$\frac{1}{4}\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Basis for $E_{\frac{1}{4}} = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$.

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Basis for $E_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

Question 2(a,b): Solution

a. $\boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$.

b. $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x - \frac{1}{4})^2(x - 1)$.

$$\tfrac{1}{4}\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Basis for $E_{\frac{1}{4}} = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$.

$$1\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{2} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Basis for $E_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

Question 2(a,b): Solution

a. $\boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$.

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Basis for $E_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

Question 2(a,b): Solution

(a) $\boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$.

(b) $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x - \frac{1}{4})^2(x - 1)$.

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$$1\boldsymbol{I} - \boldsymbol{A} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{2} \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Basis for $E_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{D}$.

Question 2(c,d): Solution

(c) $\boldsymbol{P}^{-1} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$.

For any positive integer n,

$$\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} = \boldsymbol{A}^n \boldsymbol{x}_0 = \boldsymbol{P} \begin{pmatrix} \frac{1}{4^n} & 0 & 0 \\ 0 & \frac{1}{4^n} & 0 \\ 0 & 0 & 1 \end{pmatrix} \boldsymbol{P}^{-1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2}{3}\left(\frac{1}{4^n}\right) + \frac{1}{3} \\ -\frac{1}{3}\left(\frac{1}{4^n}\right) + \frac{1}{3} \\ -\frac{1}{3}\left(\frac{1}{4^n}\right) + \frac{1}{3} \end{pmatrix}.$$

(d) As n gets large, $\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} \to \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$.

Question 2(c,d): Solution

(c) $\boldsymbol{P}^{-1} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$.

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Question 3

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A 4×3 matrix $\boldsymbol{A}$ has the following properties:

A. $\boldsymbol{A}$ has a singular value decomposition $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, with

$$\boldsymbol{V} = \begin{pmatrix} 0 & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix};$$

B. $\frac{1}{2\sqrt{5}}\boldsymbol{A}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $\frac{1}{6}\boldsymbol{A}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$, $\frac{1}{3}\boldsymbol{A}\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

C. the characteristic polynomial of $\boldsymbol{A}^T\boldsymbol{A}$ is $(x-3)(x-6)(x-10)$.

Find all possible values of $\boldsymbol{A}$.

Question 3: Solution

From (C), $\sigma_1 = \sqrt{10}$, $\sigma_2 = \sqrt{6}$ and $\sigma_3 = \sqrt{3}$, so $\boldsymbol{\Sigma} = \begin{pmatrix} \sqrt{10} & 0 & 0 \\ 0 & \sqrt{6} & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix}$.

Let $\boldsymbol{u}_i$ denote the i-th column of $\boldsymbol{U}$, and let $\boldsymbol{v}_j$ denote the j-th column of $\boldsymbol{V}$. From (B), we get

$$\boldsymbol{u}_1 = \frac{1}{\sigma_1}\boldsymbol{A}\boldsymbol{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \ \boldsymbol{u}_2 = \frac{1}{\sigma_2}\boldsymbol{A}\boldsymbol{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \ \boldsymbol{u}_3 = \frac{1}{\sigma_3}\boldsymbol{A}\boldsymbol{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 & \\ 1 & 0 & 0 & \boldsymbol{u}_4 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \\ 0 & 0 & 1 & \end{pmatrix} \begin{pmatrix} \sqrt{10} & 0 & 0 \\ 0 & \sqrt{6} & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$

$$= \begin{pmatrix} \sqrt{2} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{5} & \sqrt{5} \\ \sqrt{2} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & -1 & 1 \end{pmatrix}.$$

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$$= \begin{pmatrix} \sqrt{2} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{5} & \sqrt{5} \\ \sqrt{2} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & -1 & 1 \end{pmatrix}.$$

Question 3: Solution

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Let $\boldsymbol{u}_i$ denote the i-th column of $\boldsymbol{U}$, and let $\boldsymbol{v}_j$ denote the j-th column of $\boldsymbol{V}$. From (B), we get

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Thus $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 & \\ 1 & 0 & 0 & \boldsymbol{u}_4 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \\ 0 & 0 & 1 & \end{pmatrix} \begin{pmatrix} \sqrt{10} & 0 & 0 \\ 0 & \sqrt{6} & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$

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GOOD LUCK
IN YOUR FINALS!